

SCLERODERMA/SYSTEMIC SCLEROSIS

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LECOM

Scleroderma.....

- *Fibrosis*
- *Vasculopathy*
- *Raynauds*
- *Renal crisis*
- *PAH*
- *ILD*



OUTLINE

- Definition/Epidemiology
- Pathophysiology
- Classification and diagnosis
- Clinical Features
- Differential Diagnoses
- Treatment

Scleroderma (Ssc) - General

- *Rare, acquired multisystem autoimmune condition with abnormal skin, organs, blood vessels*
- *Peak: 35-65*
- *female: male 10:1*
- *possible etiologies: parvovirus B19, CMV, silica dust, organic solvents...*

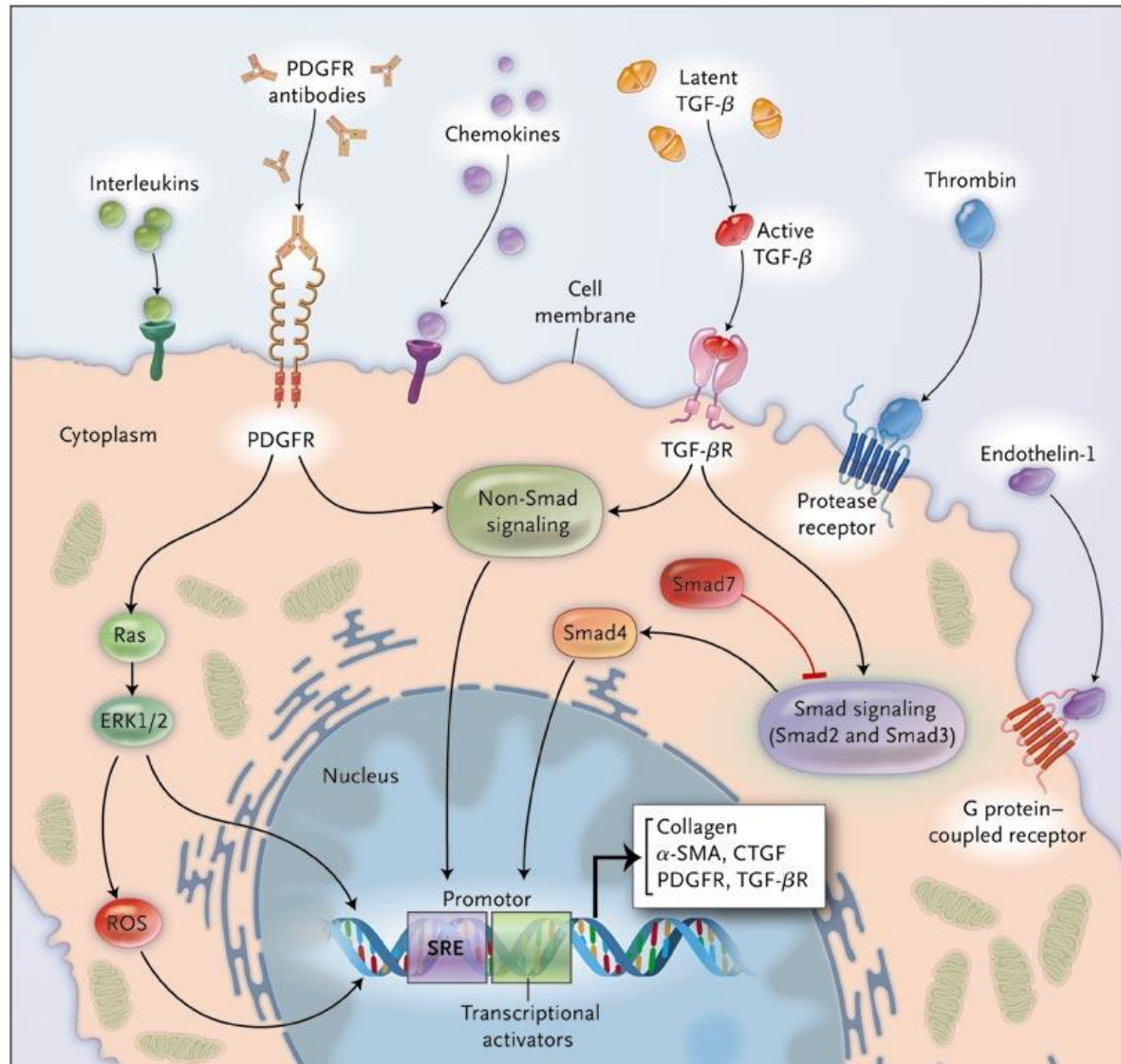
Scleroderma - Pathophysiology

- *fibroblasts*
 - *collagen deposition*
 - *IL-10 and TGF β*
- *blood vessel fibrosis/vasoconstriction*
- *proliferation of smooth muscle cells in intimal layer of blood vessels*

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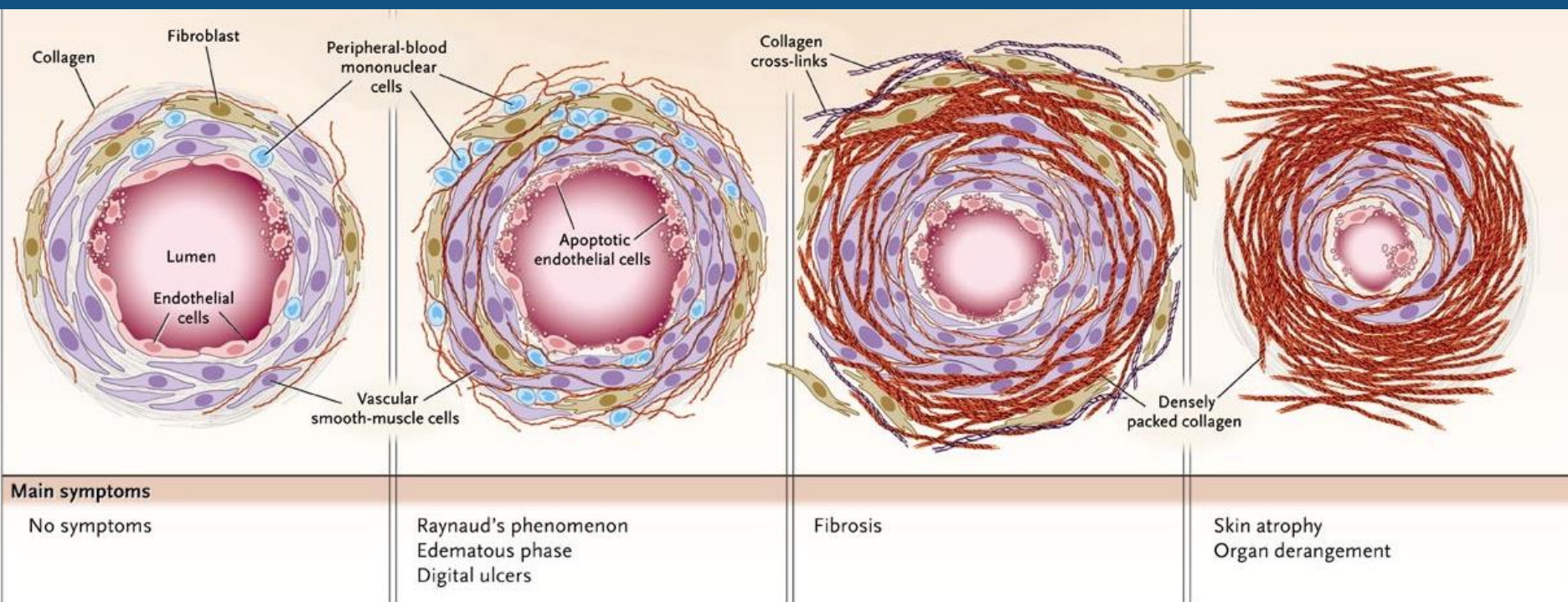
Activation of Fibroblasts in Scleroderma.



Scleroderma - Pathophysiology

- *Activation of fibroblasts and collagen deposition via IL-10 and TGF β*
- *blood vessel fibrosis/vasoconstriction*
- *proliferation of smooth muscle cells in intimal layer of blood vessels*

Lesions in Different Stages of Scleroderma.



Scleroderma:

- *Classification*

2013 ACR / EULAR Criteria For The Classification Of Systemic Sclerosis (Scleroderma)*

Item	Sub-items(s)	Weight/score [†]
Skin thickening of the fingers of both hands extending proximal to the metacarpophalangeal joints (<i>sufficient criterion</i>)	-	9
Skin thickening of the fingers (<i>only count the higher score</i>)	Puffy fingers	2
	Sclerodactyly of the fingers (distal to the metacarpophalangeal joints but proximal to the proximal interphalangeal joints)	4
Fingertip lesions (<i>only count the higher score</i>)	Digital tip ulcers	2
	Fingertip pitting scars	3
Telangiectasia	-	2
Abnormal nailfold capillaries	-	2
Pulmonary arterial hypertension and/or interstitial lung disease (<i>maximum score is 2</i>)	Pulmonary arterial hypertension	2
	Interstitial lung disease	2
Raynaud's phenomenon	-	3
SSc-related autoantibodies (anticentromere, anti-topoisomerase I [anti-Scl-70], anti-RNA polymerase III) (<i>maximum score is 3</i>)	Anticentromere 3 Anti-topoisomerase I Anti-RNA polymerase III	3

* The criteria are not applicable to patients with skin thickening sparing the fingers or to patients who have a scleroderma-like disorder that better explains their manifestations (e.g., nephrogenic sclerosing fibrosis, generalized morphea, eosinophilic fasciitis, scleredema diabeticorum, scleromyxedema, erythromyalgia, porphyria, lichen sclerosis, graft-versus-host disease, diabetic cheiroarthropathy).

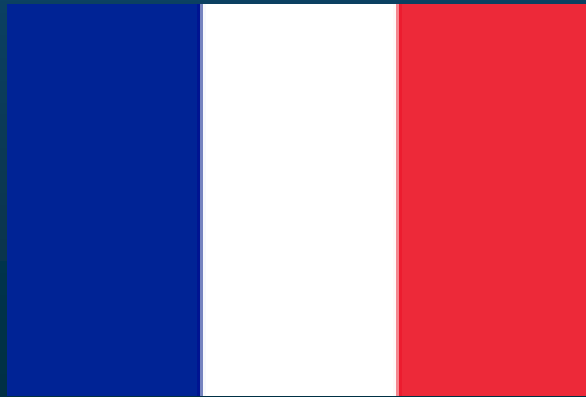
† The total score is determined by adding the maximum weight (score) in each category.
Patients with a total score of ≥ 9 are classified as having definite scleroderma.

Sensitivity 91%

Specificity 92%

Scleroderma – Clinical Features - Skin

- *Early nonspecific symptoms (years)*
- *Swelling, itch and stiffness of the hands*
- *Dilated capillary nailfold loops/Raynaud's/telangiectasiae*
- *2 subsets:*
 - *Diffuse (**anti-scl-70**/anti-topoisomerase, **RNA polymerase**)*
 - *Limited (CREST SYNDROME)(**anti-centromere**)*
 - ***calcinosis**, **raynaud's**, **e**sophageal dysmotility, **sclerodactyly**, **telangiectasia***



Maurice Raynaud (1834–1881)



Early arthritis/"puffy fingers"



Digital pits



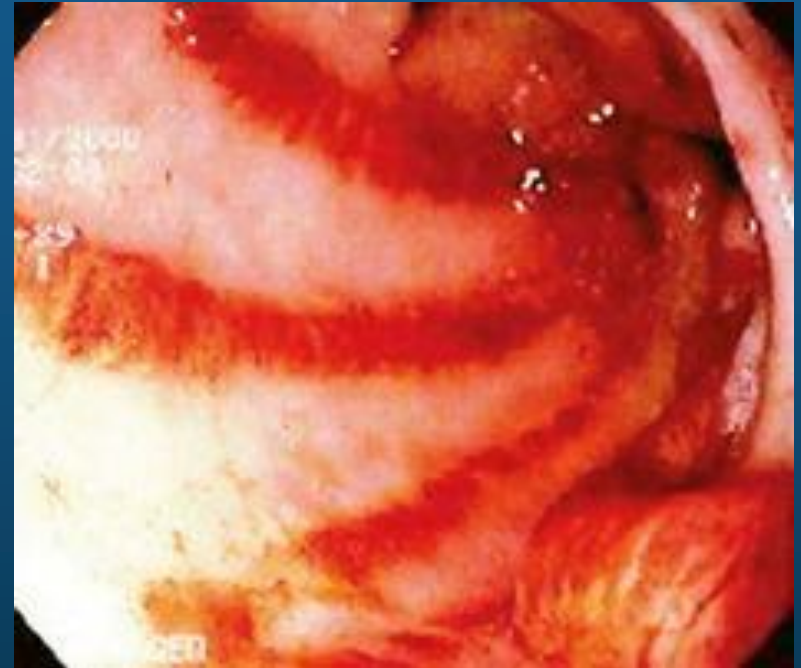
sclerodactyly



Late changes of SSC with atrophy



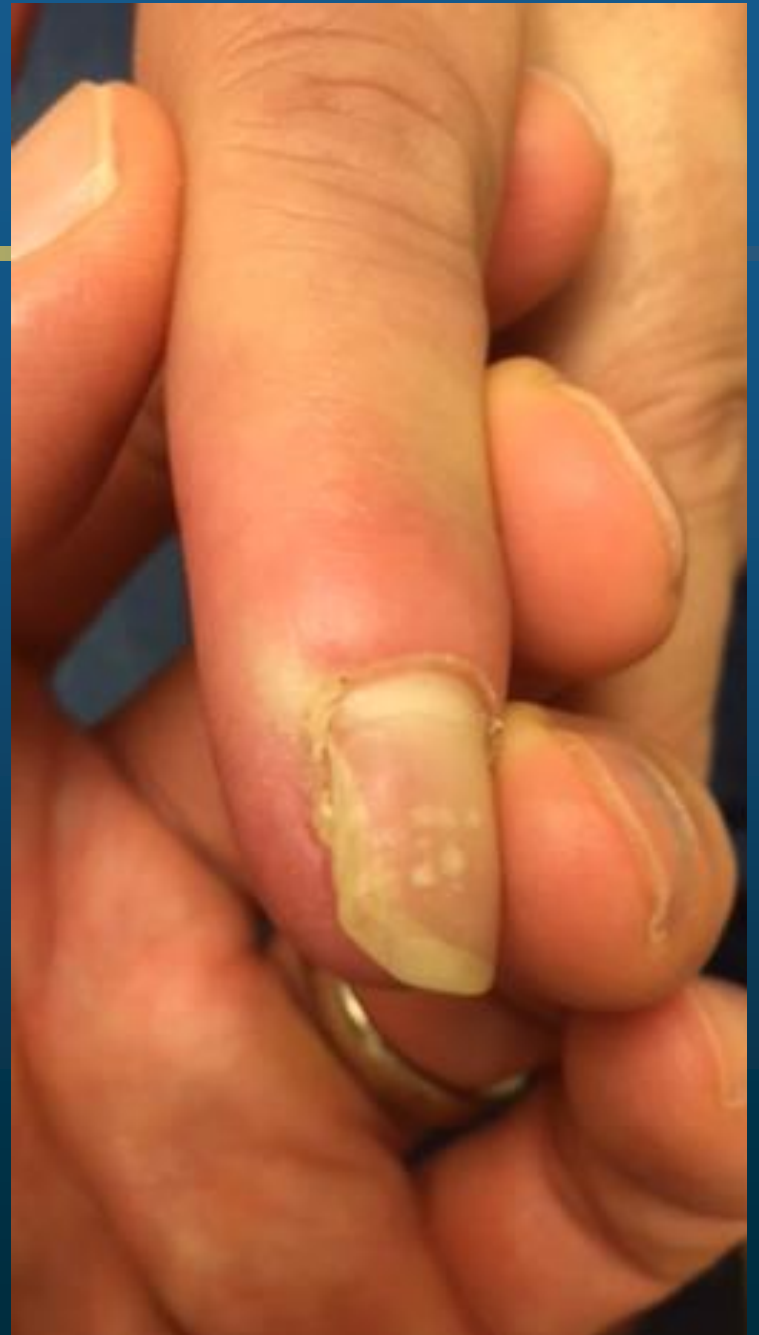
Telangiectasia



GAVE
(gastric antral vascular ectasia)
“watermelon stomach”

Raynaud's phenomenon





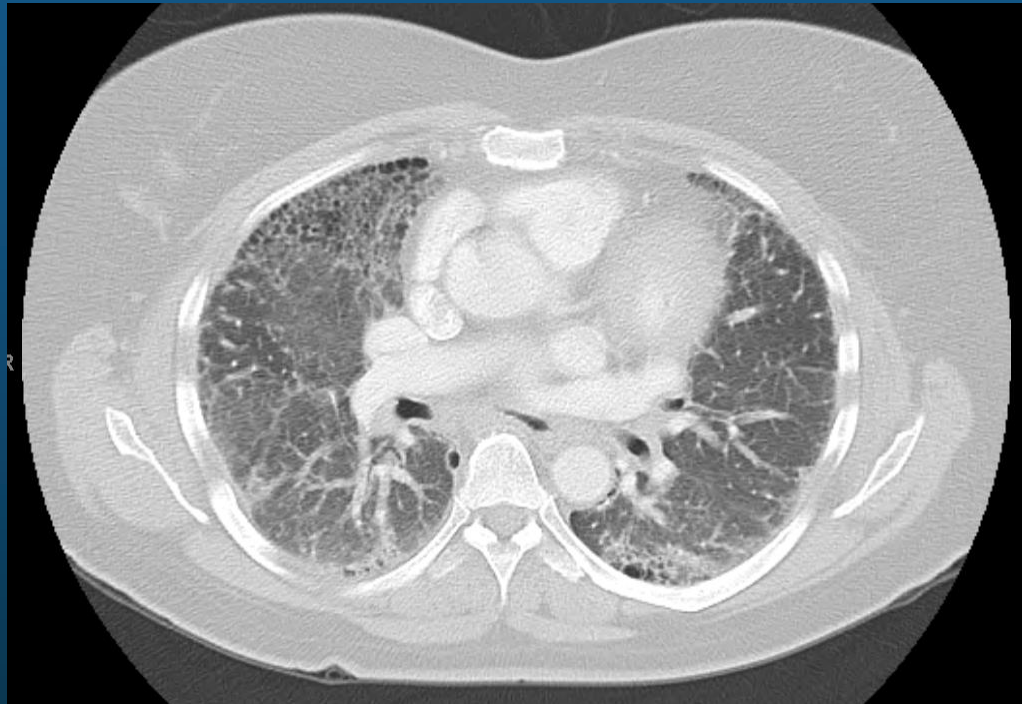
Calcinosis of CREST syndrome

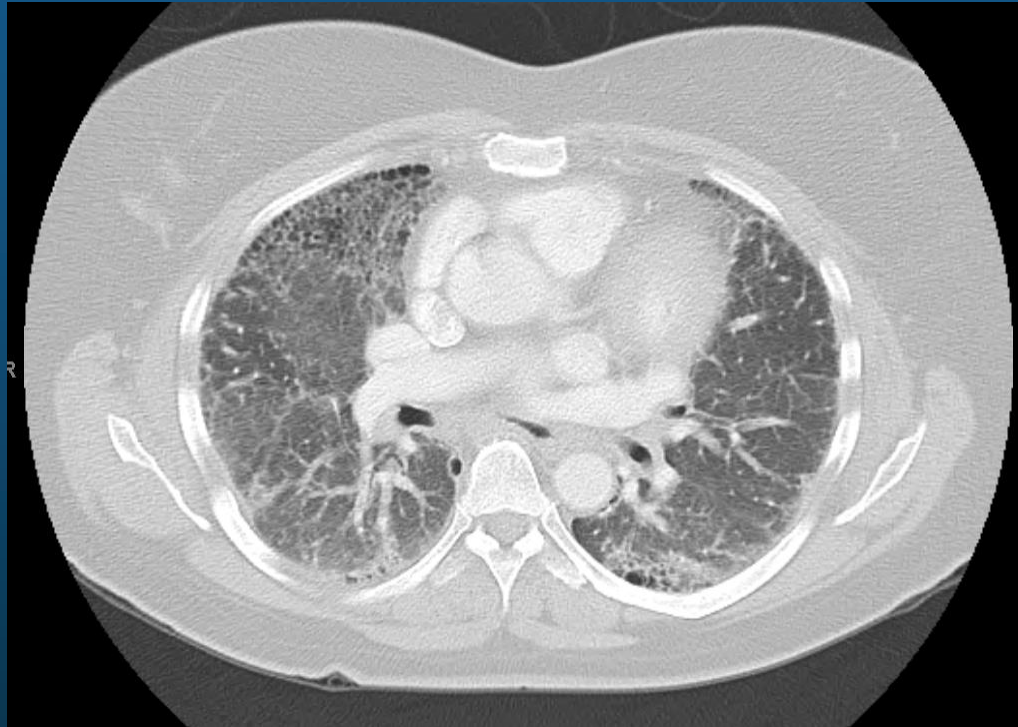


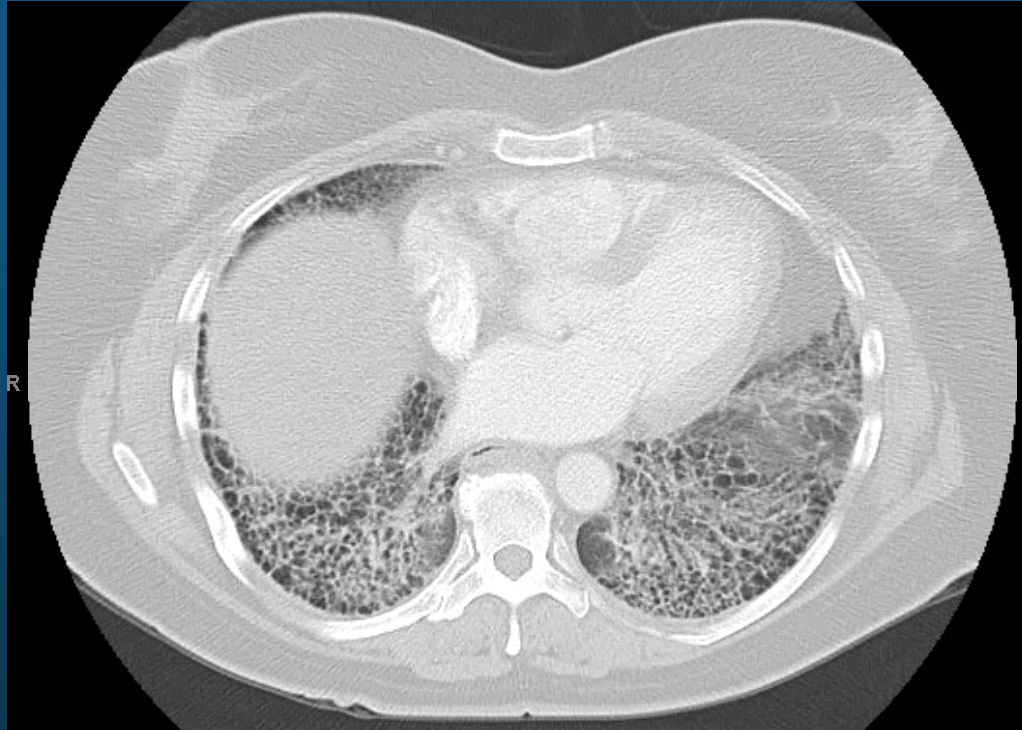
Scleroderma – Clinical Features - Lungs

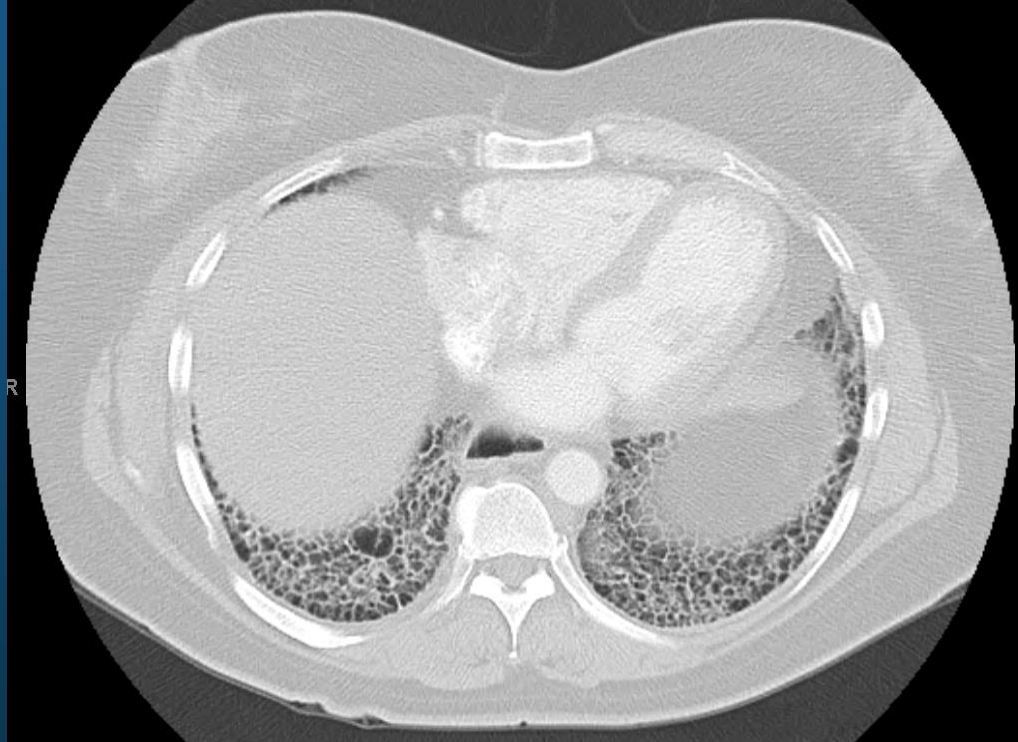
- *Non-specific interstitial pneumonia (NSIP)*
 - *alveolitis*
- *Usual interstitial pneumonitis (UIP)*
 - *fibrosis in parenchyma (honeycombing)*
- *Dyspnea, decreased lung volumes and/or diffusing capacity*
- *Pulmonary artery hypertension (PAH) with or without interstitial lung disease (ILD)*
 - *Associated with anti-nucleolar antibody*
- *Pericarditis, myocardium*
- *10 yr survival 40-70% historically*

Interstitial lung disease









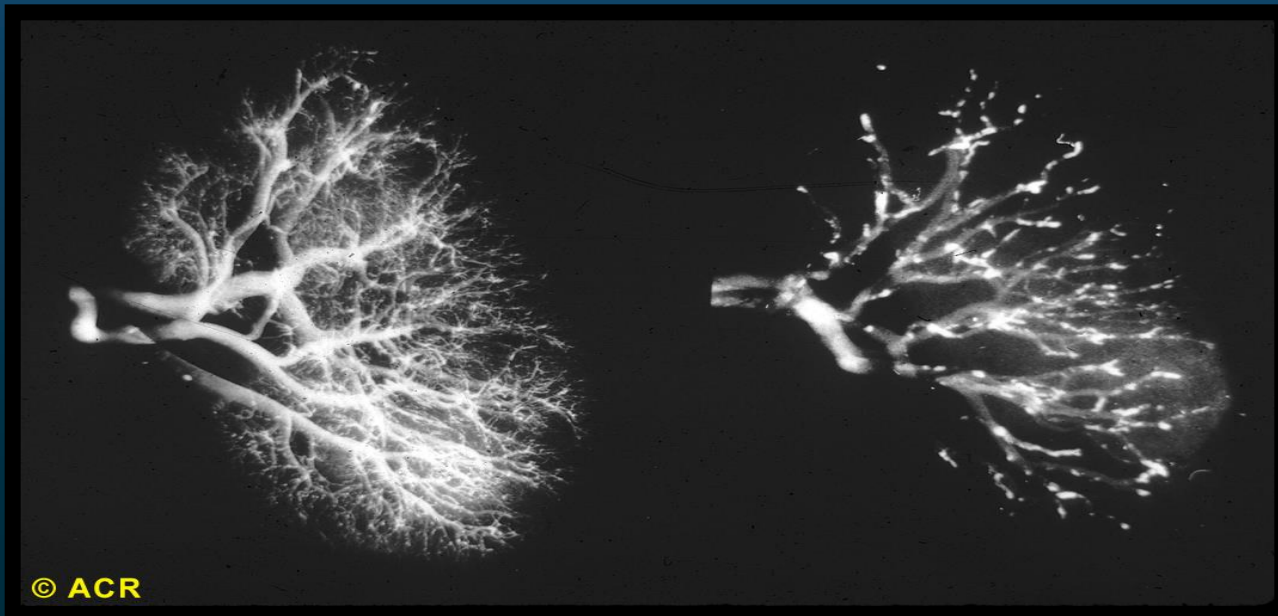
Scleroderma – Clinical Features - GI

- *commonly involved*
- *small oral aperture, dry mucosa, dysphagia, esophagitis*
- *decreased peristalsis*
- *delayed emptying of the stomach*
- *Small intestinal bacterial overgrowth (SIBO)*
 - *Achlorhydria*
 - *Pancreatic insufficiency*
 - *Microbiome*
 - *Rifaximin*



Scleroderma – Clinical Features - Renal

- *Scleroderma renal crisis (SRC)*
 - → *emergent*
 - *severe hypertension*
 - *acute renal failure*
 - *usually late manifestation of diffuse skin disease*



Scleroderma: Differential Dx

- ?

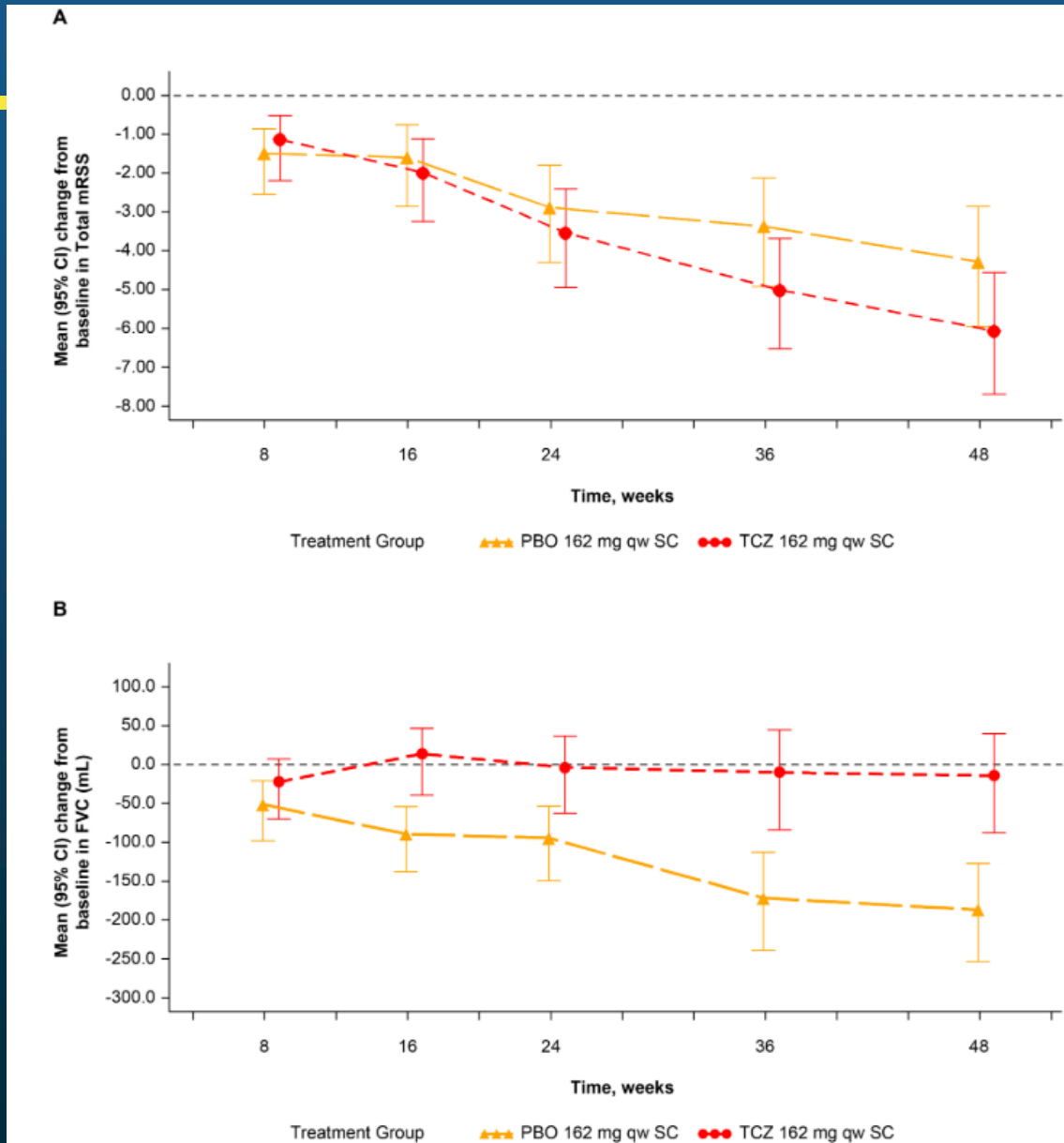
Scleroderma: Differential Dx

- *Eosinophilic fasciitis*
 - *spares the hands and face*
 - *eosinophilia on biopsy*
- *Scleredema*
 - *Idiopathic*
 - *Diabetes*
- *Nephrogenic Fibrosing Dermopathy*
- *Morphea*

Scleroderma - Treatment

- “no rheumatic medication should be discarded before a trial for scleroderma”
- penicillamine, methotrexate, chlorambucil, colchicine
- Symptomatic treatment:
 - Raynaud’s: warming measures, tobacco, calcium channel blockers, topical nitrates, botox
 - GI: PPIs, promotility agents
 - CV/pulm: cytoxan, **endothelin** antagonists (bosentan/Tracleer), **prostacyclin** analogs (epoprostenol/Flolan, iloprost/Ventavis)
 - Renal: **ACE inhibitors**
- **Immunomodulators and signal transduction?: mycophenolate mofetil/Cellcept, **rituximab**/Rituxan, imatinib/Gleevec**
- **Stem Cell Transplant; pirfenidone (Esbriet)?**
- **Nintedanib (Ofev) – sept 9, 2019 FDA approval for Scl-ILD**
- **Tocilizumab (Actemra) – march 4, 2021 FDA – Scl-ILD**

Khanna, D. Efficacy and Safety of Tocilizumab for the Treatment of Systemic Sclerosis: Results from a Phase 3 Randomized Controlled Trial. Abstract 898, ACR 2018



Scleroderma: conclusions

- *A chronic multi-system autoimmune condition of fibroblasts and B cells*
 - *Rheumatology/Pulmonary/Nephrology/Cardiology*
 - *Emerging therapies – Chimeric Antigen Receptor T cells (CAR-T)*
- *Monitoring and follow up:*
 - *Chest xrays/high resolution CT*
 - *Pulmonary function testing with DLCO*
 - *Echocardiography*
 - *Right sided heart catheterization*

THANK YOU FOR YOUR ATTENTION!
